Cross-Lingual Benchmarks for Multimodal Claim Check-Worthiness Detection in Low-Resource Languages

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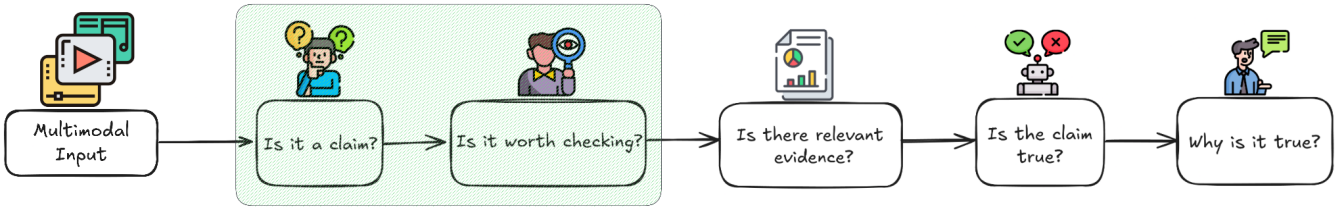


Figure 1: Automated Fact-Checking Pipeline highlighting the Claim Detection and Check-Worthiness Detection components

Abstract

The proliferation of multimodal misinformation on social media presents a critical global risk, particularly in linguistically diverse and resource-poor regions like South Asia. Despite their significant populations, languages such as Nepali, Bangla, and Telugu remain severely underrepresented in automated fact-checking research, creating a "visual blind spot" where imagery is frequently used to distort factual claims. In this paper, we introduce a novel multimodal dataset and a systematic benchmarking framework to address these geographical and modal gaps in check-worthiness detection. We evaluate over 15 state-of-the-art architectures—encompassing text-only, image-only, and multimodal Large Vision-Language Models (LVLMs)—to identify task-specific reliability across within-language and cross-lingual settings. Our results demonstrate that while zero-shot transfer from English-centric models is limited, multimodal architectures significantly outperform unimodal baselines, especially when leveraging few-shot target-language context. Our results provide the first comprehensive benchmarks for multimodal check-worthiness in these low-resource environments, highlighting the necessity of joint text-image modeling and identifying critical limitations in current cross-lingual transfer strategies.

CCS Concepts

• **Computing methodologies** → **Information extraction**.

Keywords

Multimodal Fact-Checking, Check-worthiness Detection, Low-resource NLP, Vision-Language Models, Cross-lingual Transfer.

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1 Introduction

The velocity of information dissemination in the modern digital landscape has fundamentally surpassed traditional verification methods, resulting in a surge of misinformation that now ranks among the most significant worldwide risks. To combat this, automated fact-checking pipelines have become an indispensable defense, typically structured to evaluate whether a statement is a claim, assess its check-worthiness, retrieve supporting evidence, and finally confirm its veracity. However, a significant disparity exists in the availability of these resources globally. While high-resource languages benefit from robust automated support, low-resource regions—particularly in South Asia—remain highly susceptible due to a lack of dedicated research. Despite languages like Nepali, Bangla, and Telugu having over 400 million speakers, they are significantly underrepresented in automated detection studies. This gap is further exacerbated by a "visual blind spot," as roughly 40% of social media content includes images that often contain misleading information overlooked by text-only analysis.

A fundamental challenge in addressing this multimodal misinformation is the assumption of "strong alignment" between text and visuals in current Vision-Language Models (VLMs). In reality, social media content frequently demonstrates "poor alignment," where the relationship between modalities is indirect, implicit, or contradictory. Furthermore, existing cross-lingual approaches struggle with cross-family scripts and the cultural complexities inherent to South Asian content. This study aims to systematically evaluate the effectiveness of multimodal learning for claim check-worthiness

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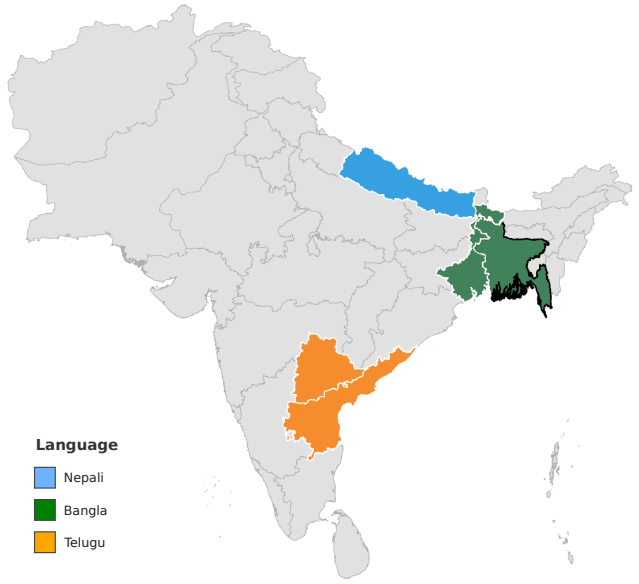


Figure 2: South-Asian map highlighting the regions with the languages represented in our dataset

detection in these under-resourced languages. We investigate the performance gains of integrating visual features over text-only baselines and analyze how well English-centric VLMs can generalize to South Asian scripts under zero-shot and few-shot conditions. Additionally, we explore cross-lingual transferability—whether models fine-tuned on one South Asian language can reduce data scarcity for others by leveraging regional commonalities. In this work, we address these critical geographical and modal gaps by establishing the first systematic benchmark for these environments. Our main contributions are as follows:

- We introduce the first high-quality multimodal dataset for claim and check-worthiness detection in Nepali, Bangla, and Telugu, consisting of 9,644 manually annotated image-text pairs.
- We establish the first comprehensive benchmarks for these low-resource environments, evaluating over 15 state-of-the-art architectures, including text-only, image-only, and Large Vision-Language Models (LVLMs).
- We provide an in-depth analysis of cross-lingual transferability, demonstrating the limitations of English-centric zero-shot transfer and the efficacy of few-shot target-language adaptation.

2 Related Work

2.1 Evolution of Automated Fact-Checking

Early research in automated fact-checking was primarily dedicated to the "triage" problem: identifying and prioritizing factual assertions from massive textual streams for human verification [22]. This foundation was established through supervised models for political discourse, most notably ClaimBuster [13], which pioneered the use of political debate transcripts for claim detection. The field has since evolved through large-scale evaluations like CLEF CheckThat!, which expanded the problem to multilingual contexts and various domains, including COVID-19 [2, 11, 21, 23]. While specialized datasets have addressed niche areas such as climate change in German tweets [16] or biomedical claims [32], these efforts share a common limitation: they rely exclusively on textual signals. This unimodal focus creates a "visual blind spot," failing to account for the increasingly multimodal nature of social media where visual elements often carry primary factual weight.

2.2 Multimodal Approaches

To bridge the modal gap, recent studies have begun integrating image analysis into the detection pipeline. Initial contributions in the multimodal domain investigated the relationship between false imagery and textual claims [33] or expanded existing text-based datasets with image content to evaluate joint detection approaches [3]. This trajectory culminated in the development of dedicated multimodal benchmarks for check-worthiness, such as the Arabic-English tweet collection by Alam et al. [1] and the specialized MM-Claims dataset [4]. Most recently, the introduction of HINTSOFTRUTH has advanced the field by providing a massive repository of over 12,000 real and synthetic claims [30]. Despite this progress, a persistent geographic bias remains; existing multimodal resources are overwhelmingly centered on high-resource languages, leaving the performance of such models in low-resource scripts largely unexplored.

2.3 Low-Resource and Cross-Lingual Settings

While multilingual research has expanded, South Asian languages remain critically under-represented in automated detection studies [10]. Foundational efforts like FactDRIL and recent Bangla-specific studies have introduced text-based verification for regional languages [26, 27], yet these lack the visual component necessary for modern social media analysis. In the absence of native-language datasets, cross-lingual transfer has emerged as a dominant strategy. Evidence suggests that multilingual transformers trained on English can provide competitive zero-shot performance in target languages [12], while entity-aware models like EX-Claim demonstrate robustness across dozens of seen and unseen scripts [24]. Nonetheless, generative Large Language Models (LLMs) still struggle with low-resource scripts compared to high-resource benchmarks [31]. This identifies a clear research gap: the need for native, multimodal benchmarks to validate the effectiveness of cross-lingual transfer in South Asian cultural and linguistic contexts.

3 Dataset

3.1 Task Definitions

Our work deals with two separate but related tasks: **Claim Detection** and **Checkworthiness Detection**. Given a text and it's corresponding image, the task is to identify whether the given image-text pair is a factual claim. If the image-text pair appears to be a factual claim, then second task is to identify whether it should be considered for fact-checking. The task definitions are derived from [1, 2]. In contrast to the existing works, we introduce a novel multilingual and multimodal dataset for claim detection

Table 1: Existing Datasets for Claim and Checkworthiness Detection

Datasets	#Samples	Modality	Language	Data Source	Task
Gupta et al., 2021[11]	9981	Text	English	Twitter	Claim Detection
Rahman et al., 2025[26]	5010	Text	Bangla	Fact-Checking Websites	Claim Detection
Iskender et al., 2021[16]	300	Text	German	Twitter	Claim, Evidence Detection
Wührl et al., 2021[32]	1200	Text	English	Twitter	Claim, Claim Type Detection
Cheema et al., 2022[4]	3400	Text, Image	English	Twitter	Claim, Checkworthiness Detection
Alam et al., 2023[1]	8045	Text, Image	Arabic, English	Twitter	Checkworthiness Detection
Meer et al., 2025[30]	12277	Text, Image	English	Mixed	Checkworthiness Detection
SATYA (Ours)	9644	Text, Image	Nepali, Bangla, Telugu	Facebook	Claim, Checkworthiness Detection

and checkworthiness detection labeled based on both the text and the corresponding image content.

3.2 Data Collection

Existing datasets for claim detection and checkworthiness detection are predominantly based on Twitter data, often limited to textual content or partially multimodal inputs, and focused primarily on high-resource languages such as English [1, 2, 11, 22]. Consequently, these resources do not adequately capture the linguistic, cultural, and contextual nuances of misinformation in low-resource regions.

To address this gap, we introduce a new dataset for three low-resource South Asian languages - Nepali, Bangla, and Telugu - covering both multimodal claim detection and checkworthiness detection tasks. We selected Facebook as our primary data source, given its extensive user base across South Asia and its widespread role in the dissemination of news and public information. Compared to Twitter, Facebook posts frequently include both textual and visual elements, offering richer contextual cues for multimodal analysis.

Data collection was conducted over a continuous period of 397 days, from March 1, 2024 to April 1, 2025, using the third-party automation platform Apify[1]. A custom scraper was configured to periodically extract each post's metadata (post ID, publication timestamp, engagement counts), the accompanying textual content (captions, descriptions, and headlines), and the corresponding images. Posts containing only videos, external links, advertisements, or other unsupported media formats were excluded.

Following data acquisition, several preprocessing and filtering steps were applied to prepare the dataset for annotation:

- *Language filtering:* Automatic language identification was used to ensure that the text content corresponded to the intended target language, removing posts that contained mixed-language or predominantly English text.
- *Text normalization:* Text was cleaned by removing escape characters, URLs, emojis, English alphabets, redundant whitespace, and orphaned punctuation. Posts with fewer than five tokens or with non-informative content (e.g., emojis or punctuation only) were removed.
- *Image filtering:* Posts containing multiple images were excluded to maintain a one-to-one mapping between text and image. Corrupted or missing images were also discarded.

[1]https://console.apify.com/actors/KoJrdxJCTtpon81KY

These steps ensured that the final dataset contained well-structured, high-quality image–text pairs in the respective local languages, providing a reliable foundation for subsequent annotation and multimodal learning experiments.

3.3 Annotation Process

To ensure high-quality and linguistically faithful annotations, all labeling was performed by native speakers of the target languages: 4 annotators for Nepali and 3 annotators each for Bangla and Telugu. All annotators had prior experience with digital media content. Before the full-scale annotation began, each annotator underwent a structured training process to familiarize themselves with the annotation interface, task definitions, and detailed instructions.

3.3.1 Annotation Framework and Interface. We built a web-based annotation interface for this study. Each post's text appeared above its image, with questions on the right and guidelines on the left for reference. The layout was clear and easy to use, helping annotators focus on the content as shown in the Appendix A.

The annotation framework followed a two-step hierarchical design based on the conceptual separation between claim detection and checkworthiness detection. The two questions presented to annotators were as follows:

Q1: "Does the image–text pair make a factual claim that can be verified?"

This question was intended to filter out purely opinion-based, humorous, or descriptive posts and to focus on posts expressing verifiable factual assertions.

Q2: "If Q1=Yes, does the image–text pair present content that is harmful, up-to-date, urgent or breaking news information or likely to mislead the public and therefore worth fact-checking?"

This question was only presented if the annotator answered "Yes" to Q1, creating a two-step decision process. The purpose was to separate claims that simply state facts from those that are significant enough to require verification by fact-checking organizations.

The questions and wording were based on previous datasets, such as MM-Claims and CheckThat! [1, 4, 22], but were simplified and adapted to be easily understood by all the annotators.

3.3.2 Training and Calibration. The annotation process consisted of three preparatory phases - Pilot, Validation, and Calibration - followed by full annotation. These steps were designed to ensure that annotators clearly understood the guidelines and applied them consistently across all posts.

Pilot Phase: Each annotator labeled a shared set of 50 image–text pairs per language. This phase served as a dry run to identify ambiguities, edge cases, or interface issues. After completing the pilot, annotators participated in a discussion to compare their decisions, clarify uncertainties, and refine the annotation guidelines with additional examples.

Validation Phase: Annotators labeled 300 new posts per language using the revised guidelines. This phase tested whether the updated instructions were consistently understood and applied. Disagreements were reviewed, particularly for borderline cases, and the guidelines were further clarified to improve uniformity.

Calibration Phase: A second set of 300 posts per language was annotated to confirm that all annotators consistently applied the finalized guidelines. This phase acted as the final preparation step before the main annotation task.

Full Annotation Phase: Following calibration, annotators labeled the remaining dataset independently. The annotation interface randomized the order of posts and concealed other annotators' responses to maintain independence. By the end of this phase, all image–text pairs were annotated according to the finalized and consistently applied guidelines.

3.4 Inter-Annotator Agreement

To assess the reliability and consistency of our annotations, we measured inter-annotator agreement (IAA) using *Krippendorff's alpha* (α) [18], which is widely used in annotation tasks due to its ability to handle multiple annotators, missing data, and nominal scales. Krippendorff's alpha provides a quantitative estimate of agreement, where values closer to 1 indicate higher reliability. Following prior work in claim and checkworthiness datasets [4], we consider values above 0.8 as *high agreement*, values between 0.67 and 0.8 as *substantial agreement*, and values between 0.5 and 0.67 as *moderate agreement*.

Multiple batches of posts were independently annotated by all annotators for each language: 16 batches for Nepali, 9 for Bangla, and 8 for Telugu. Each batch included annotations for two tasks - Claim Detection and Checkworthiness Detection. Krippendorff's alpha was computed for each batch, and the average agreement across batches is reported in Table 2.

Table 2: Average Krippendorff's alpha (α) for Claim Detection and Checkworthiness Detection across languages.

Language	Claim Detection	Checkworthiness Detection
Nepali	0.766	0.799
Bangla	0.730	0.713
Telugu	0.754	0.795

These results show *substantial agreement* across all languages and tasks, indicating that the guidelines were clearly understood and consistently applied. Gold-standard labels were established via majority consensus, and the final dataset statistics are summarized in Table 3.

Table 3: Dataset Statistics

Language	Task	Label	#Samples
Nepali	Claim	Claim	2526
		No Claim	2130
	Checkworthiness	Check-worthy	1182
		Not Check-worthy	1344
Bangla	Claim	Claim	986
		No Claim	1690
	Checkworthiness	Check-worthy	742
		Not Check-worthy	244
Telugu	Claim	Claim	554
		No Claim	1758
	Checkworthiness	Check-worthy	260
		Not Check-worthy	294

4 Experimental Models

To comprehensively evaluate claim and check-worthiness detection in low-resource South Asian languages, we benchmark a diverse suite of models across three primary categories. This experimental setup allows us to isolate the specific contribution of individual modalities and quantify the performance advantages of joint text-image modeling.

4.1 Unimodal Models

We establish unimodal baselines to determine the independent predictive power of textual and visual cues in isolation.

Text-only: We utilize widely adopted multilingual transformers, specifically XLM-RoBERTa [5], mBERT [7], mDeBERTa [15], and IndicBERT [8]. These models were selected for their robust cross-lingual representations and specialized pre-training on Indic and South Asian scripts.

Image-only: To assess whether visual features alone can signal factual claims or check-worthiness, we employ standard convolutional and transformer-based architectures: ResNet50 [14], ViT [9], and ConvNeXt [20]. These models are trained independently on post imagery to identify visual patterns associated with misinformation.

4.2 Multimodal Models

Our multimodal benchmarks integrate both discriminative encoders and generative vision-language architectures to evaluate various fusion and reasoning strategies.

Discriminative Encoders: We utilize CLIP [25] and ViLT [17] to examine fixed-feature and patch-based cross-modal alignment. These models serve as established benchmarks for discriminative multimodal classification.

Generative Vision-Language Models: To investigate the zero-shot and few-shot potential of large-scale architectures, we benchmark state-of-the-art models including Qwen2.5-VL [29], Gemma-3 [28], LLaVA-1.5 [19], and Ayavision-8B [6]. This selection allows for a direct comparison between English-centric foundations and models

with enhanced multilingual support, such as Ayavision, which is critical for the target low-resource scripts.

5 Experimental Setup

Our experiments are designed to systematically evaluate within-language performance and cross-lingual generalization across Nepali, Bangla, and Telugu.

5.1 Dataset Splits

All experiments utilize our curated multilingual dataset, split into training, validation, and test sets with ratios of 70%, 15%, and 15%, respectively, across all target languages. These consistent partitions ensure fair comparisons across all models and modalities.

5.2 Within-Language Performance (RQ1)

To evaluate baseline performance under fully supervised conditions, all text-only, image-only, and multimodal models are fine-tuned separately on the training set of each language. We utilize Parameter-Efficient Fine-Tuning (PEFT) via LoRA adapters, updating a small set of task-specific weights while keeping the majority of pretrained parameters frozen. Fine-tuned models are subsequently evaluated on the corresponding test set of the same language.

5.3 Cross-Lingual Transfer from English (RQ2)

To assess how well English-centric foundations generalize to South Asian scripts, we evaluate generative vision-language models in zero-shot and few-shot settings. Zero-shot evaluation is performed by applying base models directly to target languages without fine-tuning, while few-shot evaluation provides in-context examples ($k \in \{1, 2, 5, 10\}$) during inference to simulate low-resource adaptation.

5.4 Cross-Lingual Transfer Across South Asian Languages (RQ3)

We study the transferability of predictive knowledge between regional languages by evaluating models fine-tuned on one language (e.g., Nepali) against the test sets of others (e.g., Bangla and Telugu). This intra-South Asian generalization is tested in both zero-shot and few-shot ($k \in \{1, 2, 5, 10\}$) settings across all language-task combinations.

5.5 Training Procedure and Hyperparameters

Models are fine-tuned on identical splits to ensure stable convergence, with learning rates and hyperparameters tuned per model. To mitigate class imbalance, we employ class weights for discriminative architectures and oversampling for generative models. Table 4 summarizes the learning rates for each model category:

5.6 Evaluation Metrics

Due to the class imbalance in our dataset, we report macro F1-score as the primary evaluation metrics. Metrics are computed *separately for claim detection and checkworthiness detection* across all three languages (Nepali, Bangla, Telugu) to provide a detailed assessment of model performance per task and language.

Table 4: Learning rates for all benchmarked models

Type	Model	Learning Rate
Text	XLM-RoBERTa	2e-5
	ndicBERT	2e-5
	mBERT	2e-5
	mDeBERTa	1.5e-5
Image	ResNet50	5e-5
	ViT	5e-5
	ConvNeXt	5e-5
Multimodal	CLIP	5e-5
	ViLT	3e-5
	Gemma-3-4b-it	1e-5
	Qwen2.5-VL-3B/7B-Instruct	2e-4
	LLaVA-1.5-7B	2e-4
	Ayavision-8B	2e-4

Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Precision:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

Recall:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

F1-score:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Macro F1: The average of F1-scores across all classes:

$$\text{Macro F1} = \frac{1}{C} \sum_{c=1}^{C} F1_c \tag{5}$$

Reporting accuracy and macro F1 per task and language ensures fair and interpretable comparisons across models, modalities, and cross-lingual transfer settings, with macro F1 particularly highlighting performance on minority classes.

6 Results and Analysis

6.1 Within-Language Performance (RQ1)

Table 5 summarizes the performance across modalities. Our results indicate that while multimodal models can achieve state-of-the-art (SOTA) results—notably *AyaVision-8B* and *Qwen2.5-7B* in Nepali—specialized textual encoders like *mBERT* and *XLM-RoBERTa* remain surprisingly competitive, frequently outperforming larger generative VLMs in Bangla and Telugu. This suggests that in low-resource social media environments, explicit linguistic cues are often more reliable than visual signals, which may introduce noise due to the "visual blind spot" of poor image-text alignment. Conversely, image-only models consistently underperform (Macro F1 $\approx$ 0.50–0.60), confirming that visual features alone lack the semantic depth required for factual verification.

A clear performance hierarchy emerged across languages, with Nepali consistently yielding the highest scores, followed by Bangla and Telugu. This trend directly correlates with the dataset scale

(Table 3); the higher volume of Nepali samples ($n = 4,656$) likely provided the necessary signal for stable parameter-efficient fine-tuning (PEFT), whereas the smaller Telugu sample size ($n = 2,312$) limited the convergence of high-parameter generative models. This finding emphasizes that data volume remains a primary bottleneck for advancing multimodal check-worthiness detection in regional South Asian scripts.

Table 5: Benchmark results (Macro F1) for Claim Detection and Checkworthiness Detection across target languages. Bold values indicate the best performance per column.

Model	Claim Detection			Checkworthiness		
	NP	BN	TE	NP	BN	TE
Text-only						
XLM-RoBERTa	0.819	**0.681**	0.523	0.733	0.666	0.567
IndicBERT	0.786	0.640	0.557	0.682	0.669	0.484
mBERT	0.825	0.669	0.565	**0.744**	**0.713**	**0.617**
mDeBERTa	0.797	0.677	0.550	0.722	0.631	0.544
Image-only						
ResNet50	0.612	0.508	0.495	0.602	0.519	0.445
ViT	0.613	0.537	0.505	0.555	0.524	0.535
ConvNeXt	0.635	0.543	0.499	0.579	0.600	0.533
Multimodal						
CLIP	0.620	0.565	0.455	0.630	0.619	0.381
ViLT	0.689	0.526	0.515	0.588	0.592	0.514
Qwen2.5-3B	0.623	0.554	**0.578**	0.702	0.619	0.557
Gemma-3-4B	0.773	0.532	0.519	0.621	0.597	0.568
Qwen2.5-7B	0.832	0.660	0.547	0.698	0.595	0.407
LLaVA-1.5-7B	0.805	0.678	0.506	0.717	0.572	0.509
AyaVision-8B	**0.833**	0.666	0.513	0.642	0.594	0.533

6.2 Cross-Lingual Transfer from English (RQ2)

The zero-shot and few-shot ($k \in \{1, 2, 5, 10\}$) evaluation of English-centric Large Vision-Language Models (LVLMs) is illustrated in Figure 3. While these models possess latent cross-lingual knowledge, their out-of-the-box performance remains significantly lower than native fine-tuned baselines, highlighting a substantial "transfer gap" for South Asian scripts. We observe that In-Context Learning (ICL) dynamics are highly model-dependent; while *Qwen2.5-7B* demonstrated robust adaptation—improving Nepali Checkworthiness F1 from 0.511 to 0.586—others like *LLaVA-1.5* exhibited performance plateaus or minor degradation as k increased. This non-linear behavior suggests a high sensitivity to prompt context in low-resource scripts, where additional examples can introduce semantic noise due to the English-centric bias of base tokenizers.

In zero-shot settings, multimodal anchoring becomes critical: English-centric models rely more on visual context when their linguistic understanding of regional scripts is weak. Here, the visual modality serves as a semantic anchor, facilitating reasoning despite sub-optimal script processing. However, even the best few-shot scores (e.g., 0.619 for Qwen2.5-3B) remain significantly below fine-tuned baselines (> 0.70), underscoring the necessity of targeted regional fine-tuning for high-stakes fact-checking.

6.3 Cross-Lingual Transfer Across South Asian Languages (RQ3)

To investigate the transferability of predictive knowledge within regional contexts, we evaluated models fine-tuned on one South Asian language against the test sets of other regional languages in the dataset. Figure 4 illustrates these zero-shot and few-shot cross-lingual transfer dynamics.

Our results demonstrate that regional transfer often provides a more robust foundation than English-centric zero-shot transfer (RQ2). Specifically, models fine-tuned on linguistically or culturally proximal languages demonstrate a "warm-start" effect; for instance, *Qwen2.5-VL-7B* fine-tuned on Telugu achieved a zero-shot F1 of 0.632 on Nepali Claim Detection, which further improved to 0.699 with few-shot target-language adaptation. This suggests that the model effectively leverages regional multimodal patterns common across South Asian social media.

Multimodal models maintain a consistent advantage in cross-lingual settings, as the visual modality acts as a universal semantic layer that remains invariant to script changes. However, transfer effectiveness is asymmetrical and scale-dependent: models fine-tuned on the larger Nepali subset generalize better to Bangla and Telugu than vice-versa. These findings suggest that regional multi-task learning is a superior strategy for scaling fact-checking to the region's most under-resourced languages.

7 Discussion

The results from the SATYA benchmark reveal a significant discrepancy between model scale and regional efficacy. While LVLMs offer immense theoretical potential, their susceptibility to the "visual noise" of poorly aligned social media content often renders them less reliable than specialized, lightweight multilingual encoders for high-stakes triage. This "transfer gap" underscores a socio-technical risk: relying on English-centric foundations without native-language adaptation can create a false sense of security in South Asian misinformation defense.

However, the robust "warm-start" effect observed in cross-regional transfer (RQ3) provides a viable path forward. By leveraging the shared cultural and linguistic context of Nepali, Bangla, and Telugu, researchers can mitigate data scarcity more effectively than through English-centric paradigms alone. Future work should expand this framework to address the burgeoning threat of short-form video and incorporate additional regional dialects. Such expansions are critical to evolving SATYA from a static benchmark into a comprehensive, inclusive ecosystem for automated fact-checking across the Global South.

8 Conclusion

In this work, we addressed the critical modal and geographic disparities in automated fact-checking by introducing **SATYA**, the first systematic multimodal benchmark for Nepali, Bangla, and Telugu. Through the curation of a high-quality dataset of 9,644 manually annotated image–text pairs, we identified a persistent "visual blind spot" in low-resource misinformation detection, where poor modality alignment often degrades the performance of large-scale generative models. Our extensive benchmarking of over 15 architectures revealed that while Large Vision-Language Models (LVLMs)

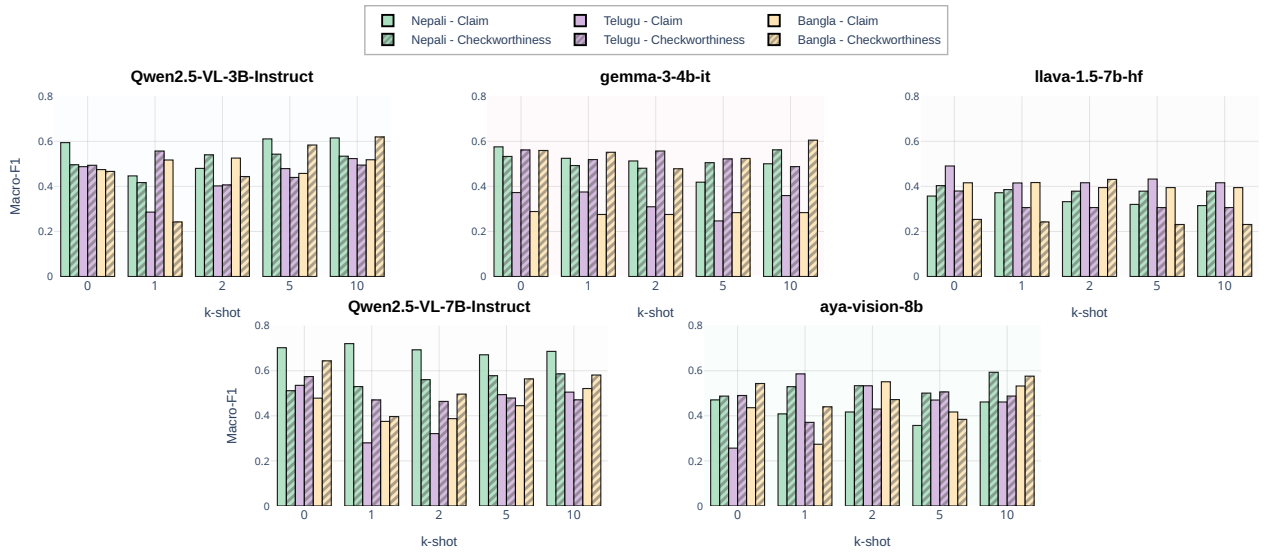


Figure 3: Macro F1 results for RQ2: Cross-lingual transfer from English base models.

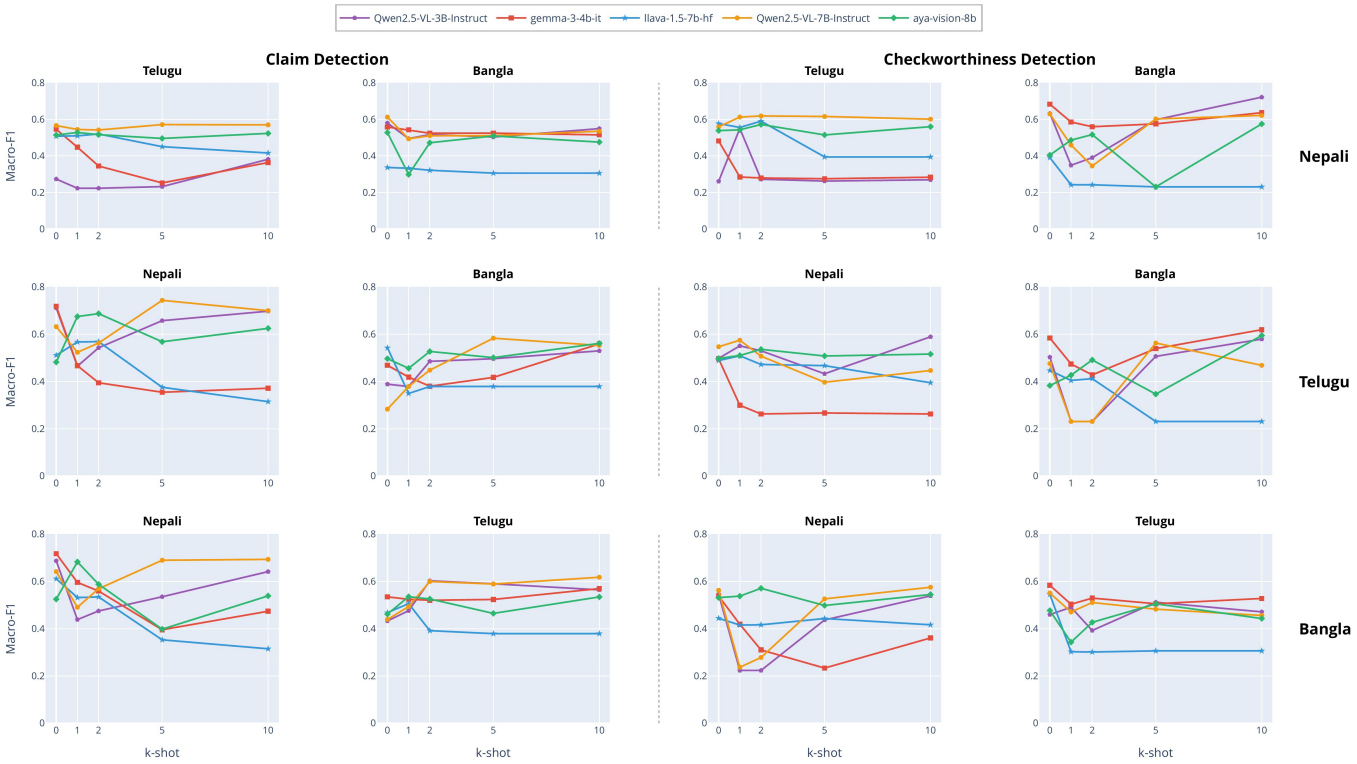


Figure 4: Macro F1 results for RQ3: Cross-lingual transfer across South Asian languages (Nepali, Bangla, and Telugu).

show promise in zero-shot settings via visual anchoring, specialized multilingual encoders remain the most robust and cost-effective solution when native-language fine-tuning is available. Furthermore, we demonstrated that regional cross-lingual transfer offers a superior alternative to English-centric paradigms, providing a sustainable "warm-start" for under-resourced scripts. By establishing these native benchmarks, we provide a foundational resource and

a methodological roadmap for building resilient, culturally-aware fact-checking systems in South Asia and beyond.

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A Annotation Interface

To ensure high-quality and consistent labeling, native speakers utilized a custom-built, web-based annotation interface. As shown in the figures below, the interface provided a structured environment

where the post's text and image were displayed alongside clear labeling guidelines and a hierarchical decision-making workflow.

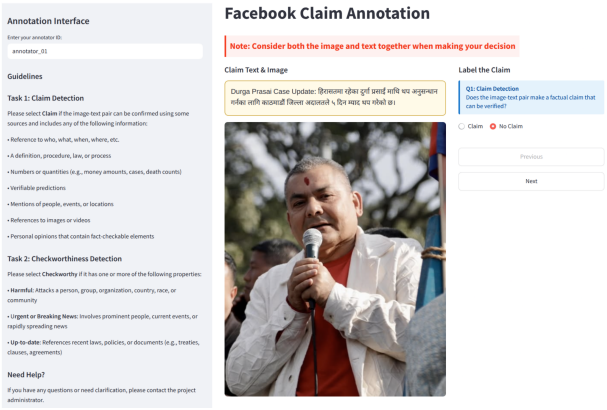


Figure 5: Primary view of the annotation interface, showing the spatial arrangement of guidelines (left), content (center), and task questions (right).

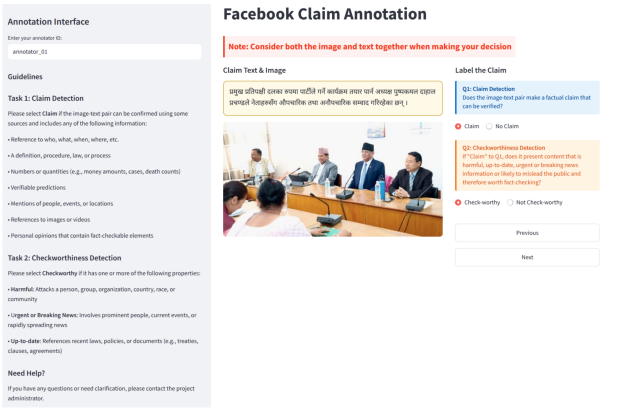


Figure 6: Detailed view of the hierarchical labeling process for Check-worthiness detection.

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